MSG No. 42.223 F_Smm2 [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `mm2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[13,14,15,16]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[17,18,19,20]`
6	$\left[0,\ \frac{1}{2},\ z\right]$	`[21,22,23,24]`
7	$\left[\frac{1}{2},\ 0,\ z\right]$	`[25,26,27,28]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[29,30,31,32]`

Table 2: Wyckoff site: `8b`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,3,22,24]`
2	$\left[0,\ \frac{3}{4},\ z\right]$	`[2,4,21,23]`
3	$\left[0,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,7,18,20]`
4	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,8,17,19]`
5	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[9,11,30,32]`
6	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[10,12,29,31]`
7	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[13,15,26,28]`
8	$\left[\frac{1}{2},\ \frac{1}{4},\ z\right]$	`[14,16,25,27]`

Table 3: Wyckoff site: `8c`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,4,26,27]`
2	$\left[\frac{3}{4},\ 0,\ z\right]$	`[2,3,25,28]`
3	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,8,30,31]`
4	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7,29,32]`
5	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	`[9,12,18,19]`
6	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	`[10,11,17,20]`
7	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	`[13,16,22,23]`
8	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[14,15,21,24]`

Table 4: Wyckoff site: `8d`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,14,24,27]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[2,13,23,28]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[3,16,22,25]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[4,15,21,26]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,10,20,31]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,9,19,32]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[7,12,18,29]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[8,11,17,30]`

Table 5: Wyckoff site: `16e`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,4]`
2	$[-x,\ 0,\ z]$	`[2,3]`
3	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,8]`
4	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{1}{2}-x,\ 0,\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[13,16]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	`[14,15]`
9	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[17,20]`
10	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[18,19]`
11	$\left[x,\ \frac{1}{2},\ z\right]$	`[21,24]`
12	$\left[-x,\ \frac{1}{2},\ z\right]$	`[22,23]`
13	$\left[x+\frac{1}{2},\ 0,\ z\right]$	`[25,28]`
14	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	`[26,27]`
15	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[29,32]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[30,31]`

Table 6: Wyckoff site: `16f`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,24]`
2	$\left[-x,\ \frac{3}{4},\ z\right]$	`[2,23]`
3	$\left[-x,\ \frac{1}{4},\ z\right]$	`[3,22]`
4	$\left[x,\ \frac{3}{4},\ z\right]$	`[4,21]`
5	$\left[x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,20]`
6	$\left[-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,19]`
7	$\left[-x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[7,18]`
8	$\left[x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[8,17]`
9	$\left[x+\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[9,32]`
10	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[10,31]`
11	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[11,30]`
12	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[12,29]`
13	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[13,28]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	`[14,27]`
15	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	`[15,26]`
16	$\left[x+\frac{1}{2},\ \frac{1}{4},\ z\right]$	`[16,25]`

Table 7: Wyckoff site: `16g`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,3]`
2	$[0,\ -y,\ z]$	`[2,4]`
3	$\left[0,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,7]`
4	$\left[0,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[6,8]`
5	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	`[9,11]`
6	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[13,15]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[14,16]`
9	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[17,19]`
10	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[18,20]`
11	$\left[0,\ y+\frac{1}{2},\ z\right]$	`[21,23]`
12	$\left[0,\ \frac{1}{2}-y,\ z\right]$	`[22,24]`
13	$\left[\frac{1}{2},\ y,\ z\right]$	`[25,27]`
14	$\left[\frac{1}{2},\ -y,\ z\right]$	`[26,28]`
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[29,31]`
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[30,32]`

Table 8: Wyckoff site: `16h`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,27]`
2	$\left[\frac{3}{4},\ -y,\ z\right]$	`[2,28]`
3	$\left[\frac{3}{4},\ y,\ z\right]$	`[3,25]`
4	$\left[\frac{1}{4},\ -y,\ z\right]$	`[4,26]`
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,31]`
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[6,32]`
7	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,29]`
8	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[8,30]`
9	$\left[\frac{3}{4},\ y,\ z+\frac{1}{2}\right]$	`[9,19]`
10	$\left[\frac{1}{4},\ -y,\ z+\frac{1}{2}\right]$	`[10,20]`
11	$\left[\frac{1}{4},\ y,\ z+\frac{1}{2}\right]$	`[11,17]`
12	$\left[\frac{3}{4},\ -y,\ z+\frac{1}{2}\right]$	`[12,18]`
13	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z\right]$	`[13,23]`
14	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[14,24]`
15	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	`[15,21]`
16	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	`[16,22]`

Table 9: Wyckoff site: `32i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ y,\ z]$	`[3]`
4	$[x,\ -y,\ z]$	`[4]`

continued ...

Table 9

No.	position	mapping
5	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5]
6	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[6]
7	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[7]
8	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[11]
12	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[14]
15	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[15]
16	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[16]
17	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[17]
18	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[18]
19	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[x,\, y+\frac{1}{2},\, z\right]$	[21]
22	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[22]
23	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[23]
24	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[24]
25	$\left[x+\frac{1}{2},\, y,\, z\right]$	[25]
26	$\left[\frac{1}{2}-x,\, -y,\, z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[27]
28	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[28]
29	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[29]
30	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[31]
32	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[32]